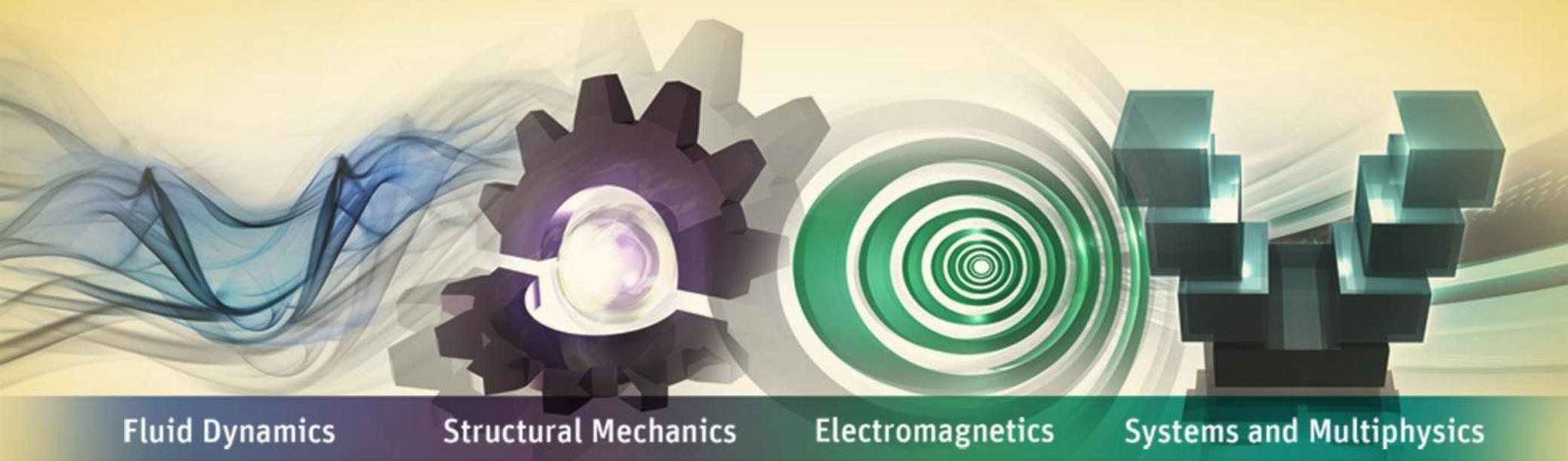


Lecture 2: General Procedure



ANSYS Mechanical

A. Building a nonlinear model

What is different about building a nonlinear model vs. a linear model?

In some cases, there will be *no* difference!

- A model undergoing mildly nonlinear behavior due to large deflection and stress stiffening effects might need no modification with regards to geometry set up and meshing.

In other cases, you must include special features:

- Elements with special properties (such as contact elements)
- Nonlinear Material data (such as plasticity and creep data)
- Include geometric features (i.e. radius at sharp corners) to overcome singularities that cause convergence trouble.
- You might also need to give special attention to:
 - Mesh control considerations under large deflection
 - Element technology options under large deflection with nonlinear materials
 - Load and boundary condition limitations under large deflection

... Building a nonlinear model

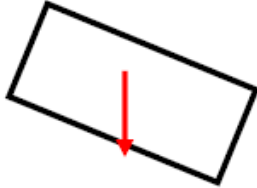
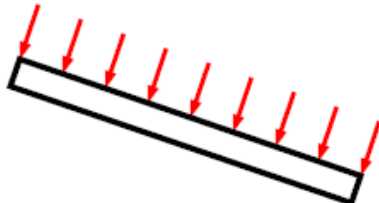
With regards to meshing, if *large strains* are expected, the shape checking option may be changed to “Aggressive”

- For large-deflection analyses, if elements undergo some change in shape, this may reduce the fidelity of the solution.
- “Aggressive” shape checking offers an improved element quality in anticipation of excessive distortion in a large-strain analysis.
- The default “Standard” shape checking is suitable for linear analyses.
- Depending on complexity of geometry, can sometimes cause failures during mesh generation
- Refer to *Mechanical – Intro, pt1* for ways to detect and remedy mesh failures.

Details of "Mesh"	
[-] Defaults	
Physics Preference	Mechanical
Relevance	0
[-] Advanced	
Relevance Center	Coarse
Element Size	Default
Shape Checking	Aggressive Mechanical
Solid Element Midside Nodes	Standard Mechanical
Straight Sided Elements	Aggressive Mechanical
Initial Size Seed	CFD
Smoothing	Explicit
Transition	Fast
[-] Statistics	

... Building a nonlinear model

It is important to note the orientation of loads and its effect on the structure in large-deflection analyses:

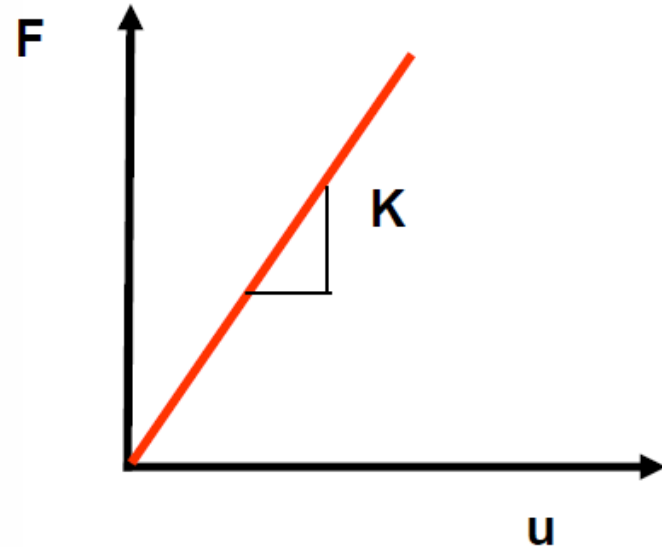
Load	Direction Before Deflection	Direction After Deflection
Acceleration (constant direction)		
Force, Moment, Bolt Load (constant direction)		
Pressure (always normal to surface)		

B. Obtaining a nonlinear solution

What is different about obtaining a nonlinear solution?

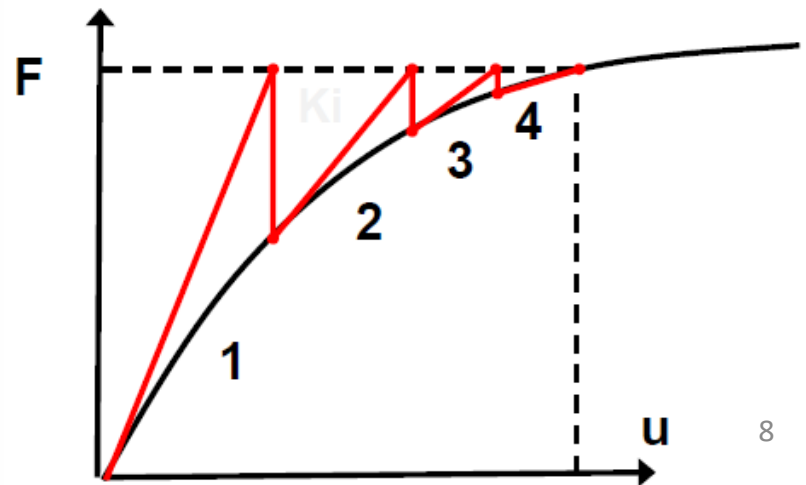
- Linear static requires only one pass through the matrix equation solver

$$F = Ku$$



- Nonlinear performs a new solution with every iteration

$$F_i = K_i u_i$$

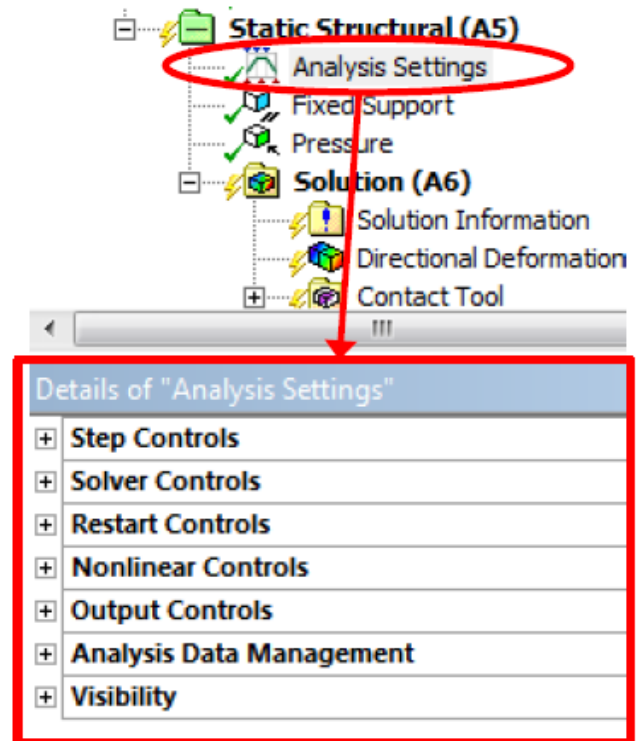


... Obtaining a nonlinear solution

...What is different about obtaining a nonlinear solution?

Analysis Settings has many options that need to be considered for a nonlinear run.

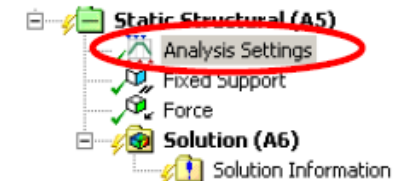
- Step Control - Load steps, Substeps, Autotime stepping
- Solver Control - Choosing the right Solver type
- Restart Controls - resuming a solve
- Nonlinear Controls - N-R convergence criteria
- Output Controls - controlling what data is saved
- Analysis Data Management – deleting/keeping files



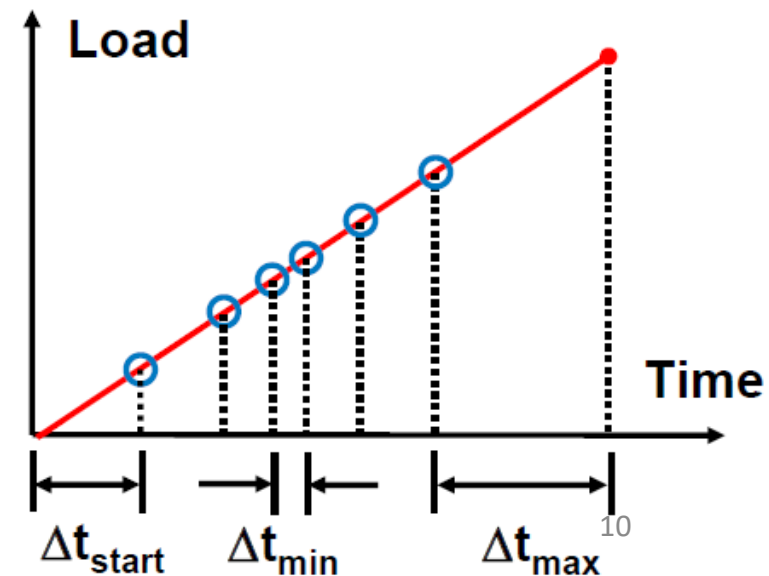
In the following slides, we consider each of these tools

Step Controls

- “Auto Time Stepping” calculates an optimum time step at the end of each substep, based on the response of the structure to the applied loads.
 - User specifies an initial number of substeps along with a range (minimum and maximum).
- Auto Time Stepping has the effect of adjusting the load increment (up *and* down) throughout the solution.
 - Smaller increments when convergence is difficult, larger increments when convergence is easy.



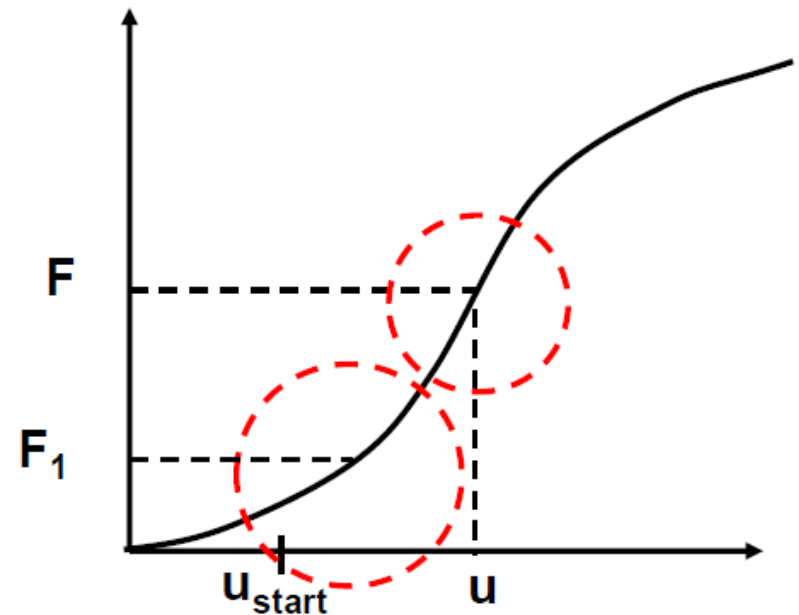
Details of "Analysis Settings"	
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	10.
Minimum Substeps	5.
Maximum Substeps	100.



... Obtaining a nonlinear solution

Step Controls (cont'd)

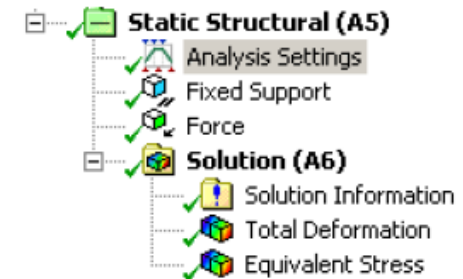
- Recall that breaking the load into increments improves convergence by bringing the start point within the radius of convergence.
- If *Mechanical* has trouble converging, the auto time stepping algorithm will *bisect* the solution.
 - “Bisection” returns to the last successfully converged substep and applies the load in a smaller increment (thereby using more substeps within the specified range).



... Obtaining a nonlinear solution

Solver Controls

- Solver Type offers two options, 'Direct' and 'Iterative'.
 - This is a reference to the way the code builds the stiffness matrix for each Newton-Raphson equilibrium iteration.
 - Direct (Sparse) solver is more robust and is recommended for challenging nonlinear models and with non-continuum elements (shells and beams).
 - Iterative (PCG) solver is more efficient (in terms of run time and is recommended for large bulk solid models dominated by linear elastic behavior.
 - There can also be different 'In-Core' memory requirements:
 - Sparse requires about 10GB/MDOF, while PCG requires 1GB/MDOF
 - The default 'Program Controlled' will automatically select a solver based on the problem currently in session.



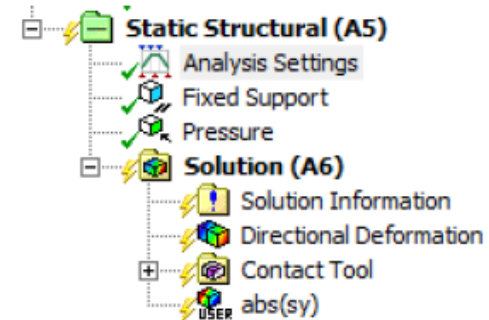
Details of "Analysis Settings"

+ Step Controls	
- Solver Controls	
Solver Type	Program Controlled
Weak Springs	Program Controlled
Large Deflection	Direct
Inertia Relief	Off

... Obtaining a nonlinear solution

Solver Controls (cont'd)

- By setting "Large Deflection" = ON, in the Solver Control branch of Analysis Settings, Adjustments are made to the stiffness matrix over multiple iterations to account for changes due to:
 - Large deflection, large rotation
 - Large strain.
 - Stress stiffening
 - Spin softening



Details of "Analysis Settings"

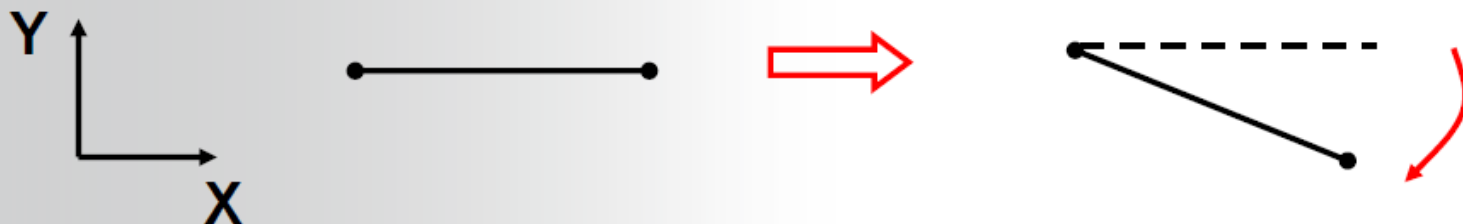
+ Step Controls	
- Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Large Deflection	On
Inertia Relief	Off
	On
+ Restart Controls	
+ Nonlinear Controls	
+ Output Controls	
+ Analysis Data Management	
+ Visibility	

... Obtaining a nonlinear solution

Solver Controls (cont'd):

Large Deflection: If an element's *orientation* changes (rotation), the transformation of its local stiffness into global components will change.

- Change in orientation is also accounted for when *Large Deflection* analysis is specified.
- It is possible to have large deflection without large strain.
- For elements that support large deflection but do not support large strain (previous slide), the mechanical strains are evaluated using *linear* expressions.

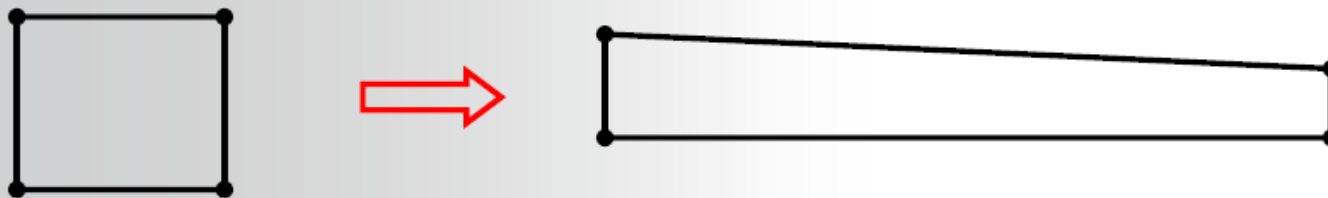


... Obtaining a nonlinear solution

Solver Controls (cont'd):

Large Deflection = On will also account for Large Strain.

If an element's *shape* changes (area, thickness, etc.), its individual element stiffness will change.



- The mechanical strains are evaluated using *nonlinear* expressions

$$\varepsilon_l = \int_{l_0}^l \frac{dl}{l} = \ln\left(\frac{l}{l_0}\right)$$

- This measure is a nonlinear strain measure since it is a nonlinear function of the unknown final length l . It is also referred to as the **log** strain. The 3-D equivalent of the log strain is the **Hencky** strain.

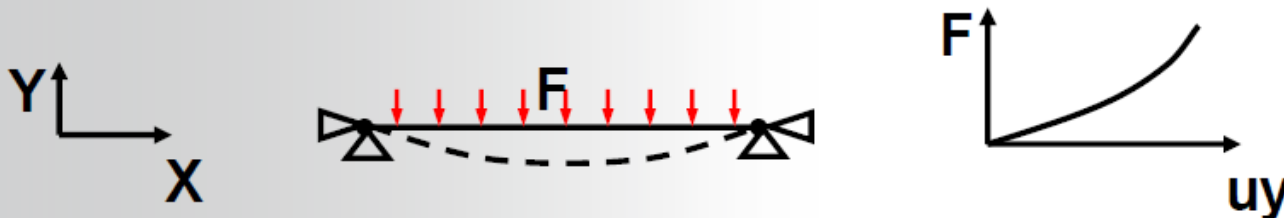
... Obtaining a nonlinear solution

Solver Controls (cont'd):

Large Deflection = On will also account for Stress Stiffening.

If an element's strains produce an in-plane stress state (*membrane stresses*), the out-of-plane stiffness can be significantly affected.

- As the vertical deflection (U_Y), increases, significant membrane stresses (S_X) lead to a stiffening response.
- This is characteristic of thin structures with bending stiffness very small compared to axial stiffness
 - cables, thin beams, shells and couples in the in-plane and transverse displacements
- It is possible to have stress stiffening without large deflection and/or large strain



Workshop 2A – Large Deflection

Please refer to your *Workshop Supplement* for instructions on:
WorkShop 2A- Large Deflection Analysis

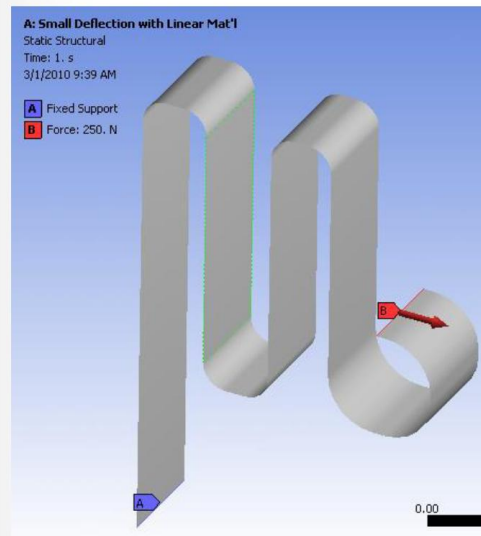
Workshop 2A: Large Deflection

Goal

Compare and contrast results using small deflection theory and large deflection theory on a model with identical loads and boundary conditions.

Model Description

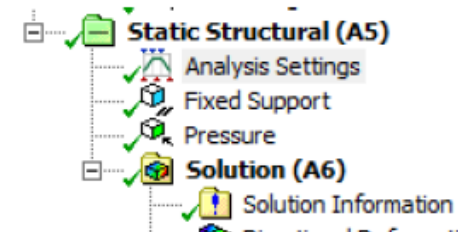
- 3D Spring plate
- Linear steel material
- Meshed with 3D Shell elements
- Fixed support at one end, A
- 250N load at opposite end, B



Obtaining a nonlinear solution

Restart Controls facilitate...

- Pausing or stopping a run to review results in progress.
- Changing analysis settings to correct an unconverged solution.
- Modifying existing Loads.
- Extending a solution that has already completed.
 - For example, to allow system transients to progress further into time.
- Adding post processing command object(s) after the model has been fully solved.



Details of "Analysis Settings"

+ Step Controls	
- Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Large Deflection	On
Inertia Relief	Off
- Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
+ Nonlinear Controls	
+ Output Controls	
+ Analysis Data Management	
+ Visibility	

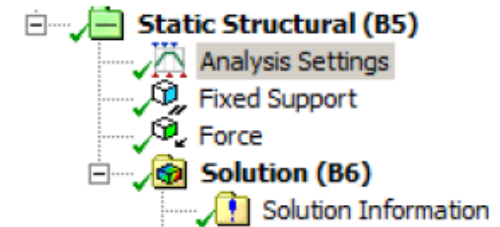
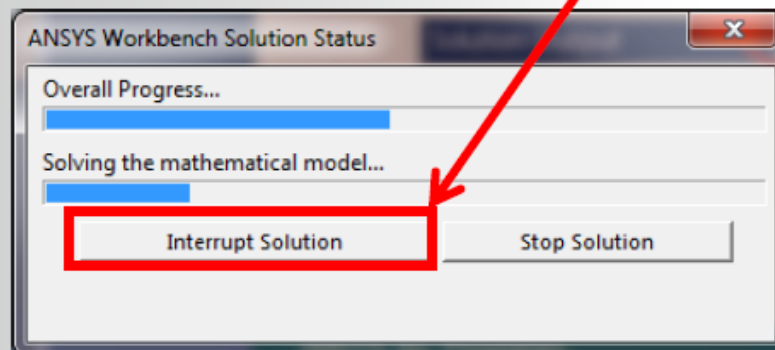
... Obtaining a nonlinear solution

Restart Controls (cont'd)

With “Generate Restart Points” set to “Program Controlled”

Restart files are...

- Not saved for a full stand alone linear analysis.
- Not saved for a full stand alone nonlinear structural analysis that completes successfully.
- Retained for the last successfully converged substep of an incomplete solve due to a convergence failure or if the solution run is manually interrupted.



Details of "Analysis Settings"

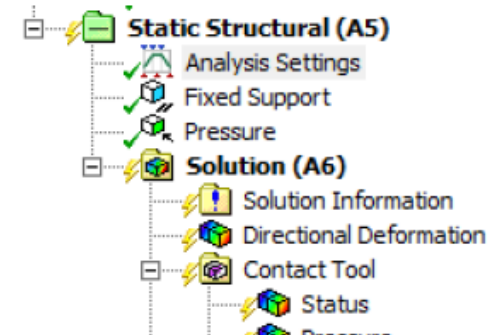
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	Program Controlled
Nonlinear Controls	Manual
Output Controls	Off
Analysis Data Management	
Visibility	

... Obtaining a nonlinear solution

Restart Controls (Cont'd)

With “Generate Restart Points” set to “Manual”...

- Load Step: Specifies at what load steps to create restart points (Last or All).
- Substep: Specifies how often the restart points are created within a load step.
 - Last: Create a restart point for the last substep of each load step only.
 - All: Creates restart points for all substeps of each load step.
 - Specified: Creates restart points for a user specified number (N) of substeps per load step.
 - Where N is defined in “Rate of Recurrence” Field
 - Equally Spaced: Creates specified number (N) of restart points at equally spaced time intervals within a load step.
 - Where N is defined in “Rate of Recurrence” Field



Details of "Analysis Settings"

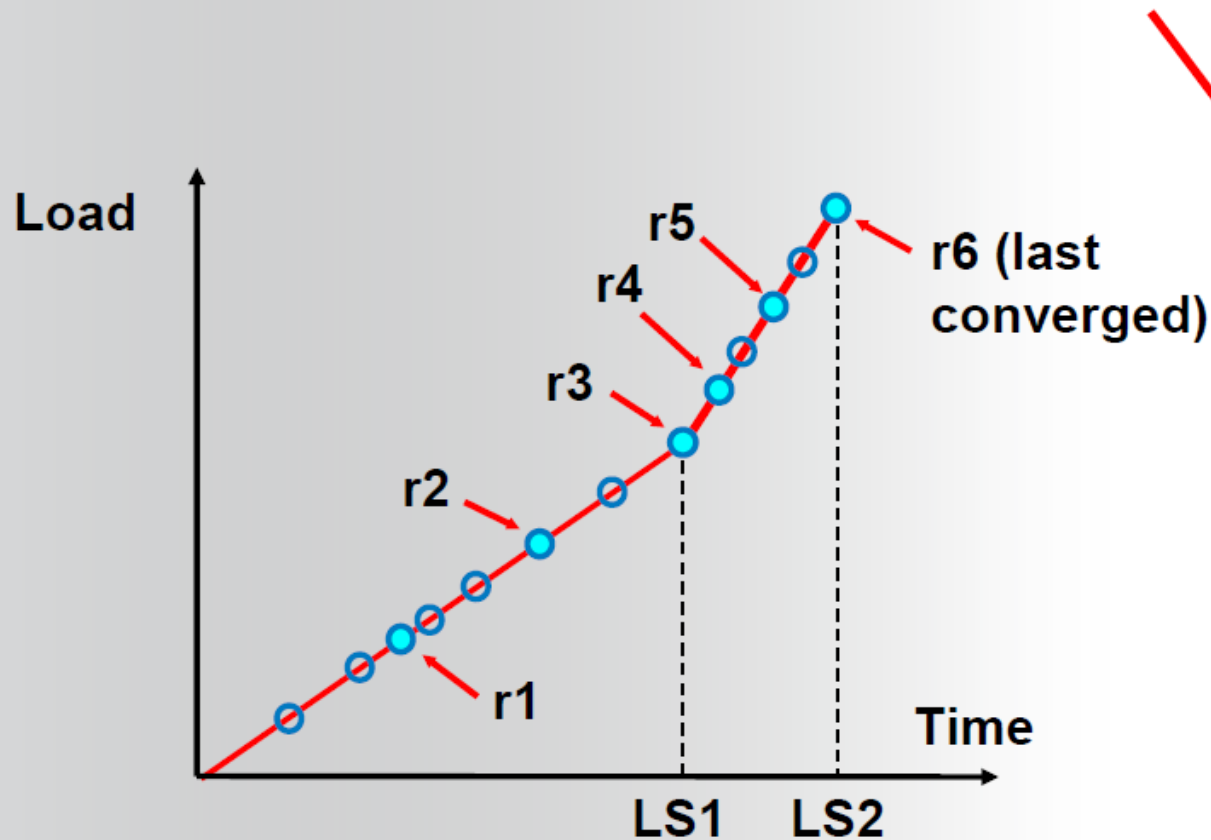
Restart Analysis	
Step Controls	
Solver Controls	
Restart Controls	
Generate Restart Points	Manual
Load Step	All
Substep	Specified
Rate of Recurrence	Last
Maximum Points to Save Per Step	All
Retain Files After Full Solve	Specified
	Equally Spaced

Restart Controls	
Generate Restart Points	Manual
Load Step	All
Substep	Specified
Rate of Recurrence	3
Maximum Points to Save Per Step	All 21
Retain Files After Full Solve	Yes

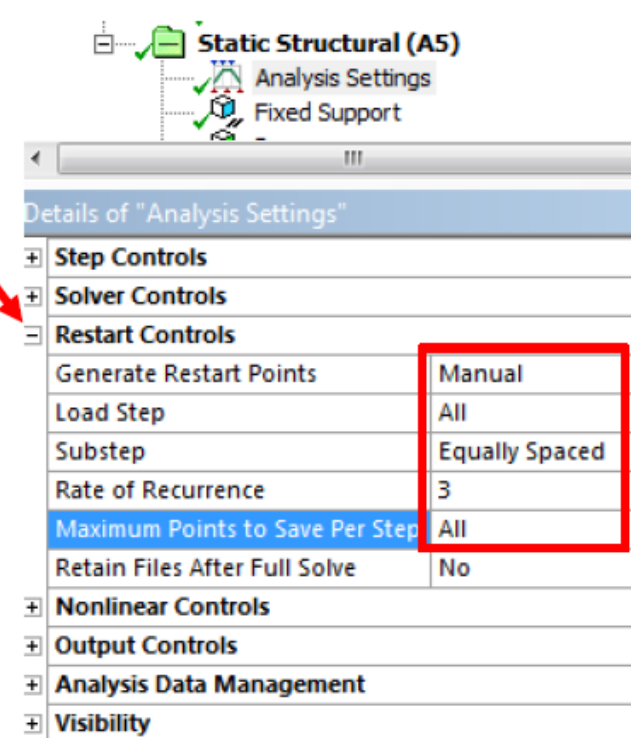
... Obtaining a nonlinear solution

Restart Controls (Cont'd)

For example, to write 3 equally spaced restart files for each load step:



- Substeps
- Restart points



... Obtaining a nonlinear solution

Restart Controls (cont'd)

Restart files are automatically deleted if a full solve completes successfully (default)

- **Retain Files After Full Solve:**
 - User has the option to keep restart files regardless by setting this field to YES.
- Under Analysis Data Management, setting Future Analysis to “Prestressed analysis” also forces the restart files to be retained.
- Similarly, setting Delete Unneeded Files to “No” implies that restart files are to be retained.

Details of "Analysis Settings"	
+ Step Controls	
+ Solver Controls	
- Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	Yes
+ Nonlinear Controls	Yes
+ Output Controls	No

Details of "Analysis Settings"	
+ Step Controls	
+ Solver Controls	
+ Restart Controls	
+ Nonlinear Controls	
+ Output Controls	
- Analysis Data Management	
Solver Files Directory	D:\TRAINING\130\ANSYS-I
Future Analysis	Prestressed analysis
Scratch Solver Files Directory	
Save MAPDL db	No
Delete Unneeded Files	No
Nonlinear Solution	Yes

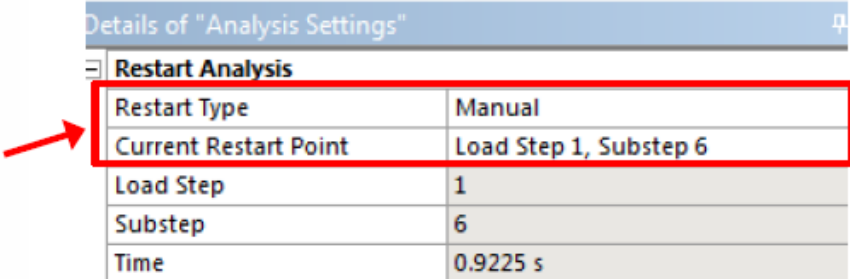
... Obtaining a nonlinear solution

Restart Controls (cont'd)

- At the completion of the run, users can specify the restart point for the subsequent run.
- If default restart controls were taken, restart will only be available for the last successfully converged substep

Restart specifications:

- Restart Type = Manual
 - Restart Point = Load Step 1, Substep 6
- Once the restart specifications have been set and the analysis control settings and/or existing loads have been adjusted as needed, execute a solve to begin the solution restart...



Restart Analysis	
Restart Type	Manual
Current Restart Point	Load Step 1, Substep 6
Load Step	1
Substep	6
Time	0.9225 s

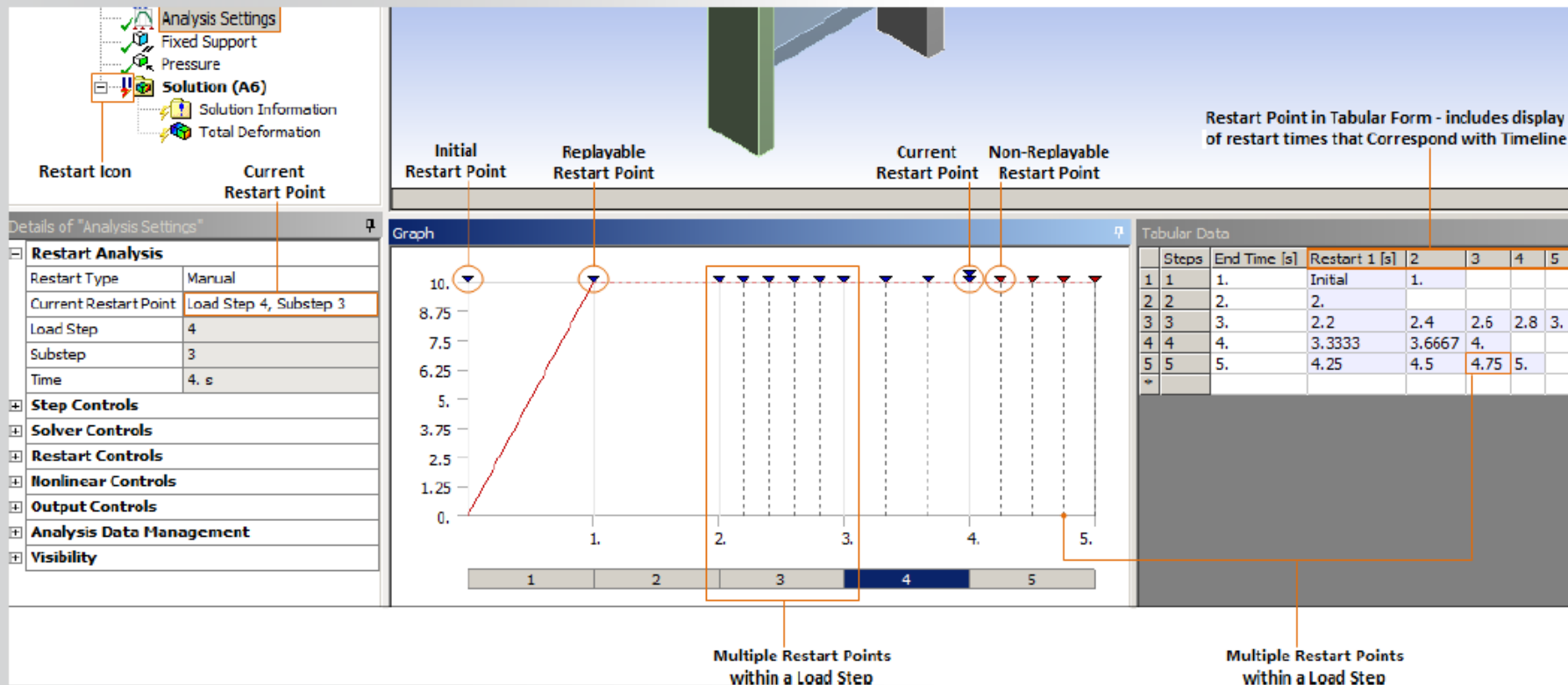


... Obtaining a nonlinear solution

Restart Controls (cont'd)

For an overview of available restart points, select the Analysis Settings object and refer to the Graph window where restart points are symbolized by triangular markers atop the timeline.

The Tabular Data window lists the restart points within each load step.

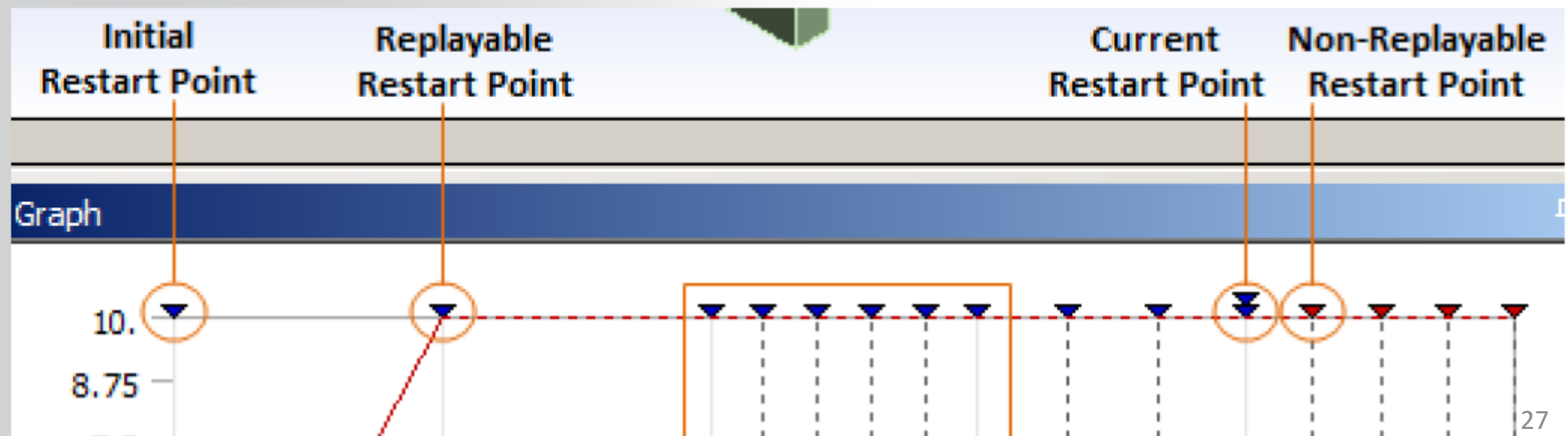


... Obtaining a nonlinear solution

Restart Controls (cont'd)

Restart points are color coded to distinguish between replayable (blue triangles) and a non-replayable (red triangles).

- Replayable points are ones which will produce the exact solution when run from start to finish.
- Non-replayable points appear if you modify a load or analysis for a given step and restart from that step. The restart will solve, but if you later solve without restarts, keeping the modifications for the entire step, non-replayable points are those which might not be available.



... Obtaining a nonlinear solution

Restart Controls (cont'd)

Below is a summary of loads supported for restarts

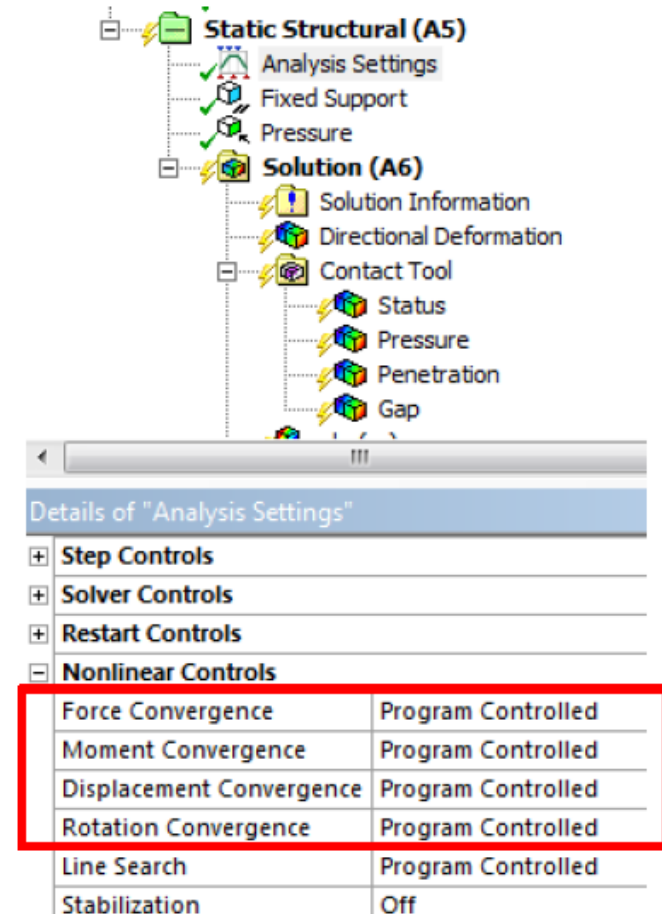
- Loads must already exist in the Project Tree from the start of the analysis
- Adding a new load into the project tree will nullify a restart

Load Type	Load Specified As...		
	Constant	Tabular	Function
Pressure	X	X	X
Line Pressure	X	X	X
Force	X	X	X
Remote Force	X	X	X
Moment	X	X	X
Displacement	X	X	N/A
Remote Displacement	X	X	N/A
Rotational Velocity	X	X	X
Bolt Pretension	X	X	N/A
Acceleration	X	X	X
Earth Gravity	N/A	N/A	N/A
Hydrostatic Pressure	X	N/A	N/A
Bearing Load	X	X	N/A
Joint Load	X	X	X
Pipe Temperature	X	X	X
Pipe Pressure	X	X	X
Thermal Condition	X	X	X
Imported Load	N/A	N/A	N/A
Nodal Force	X	X	X
Nodal Pressure	X	X	X
Nodal Displacement	X	X	N/A

... Obtaining a nonlinear solution

Nonlinear Controls

- Tolerances on Convergence are calculated automatically. They are used during the Newton-Raphson process to dictate when a model is Converged or “balanced”
 - The default convergence criterion works very well for most engineering applications.
 - For special situations, users can override these defaults to Tighten or loosen the convergence tolerance.
 - A tighter tolerance gives better accuracy, but can make convergence more challenging



The screenshot displays the ANSYS Workbench interface. The top part shows the 'Static Structural (A5)' analysis tree, which includes 'Analysis Settings', 'Fixed Support', 'Pressure', and 'Solution (A6)'. The 'Solution (A6)' branch is expanded, showing 'Solution Information', 'Directional Deformation', 'Contact Tool', 'Status', 'Pressure', 'Penetration', and 'Gap'. Below the tree, the 'Details of "Analysis Settings"' panel is visible. It contains several expandable sections: 'Step Controls', 'Solver Controls', 'Restart Controls', and 'Nonlinear Controls'. The 'Nonlinear Controls' section is expanded, revealing a table of convergence settings. The table has two columns: the first column lists the convergence criteria, and the second column shows the current setting. The first four rows of the table are highlighted with a red border.

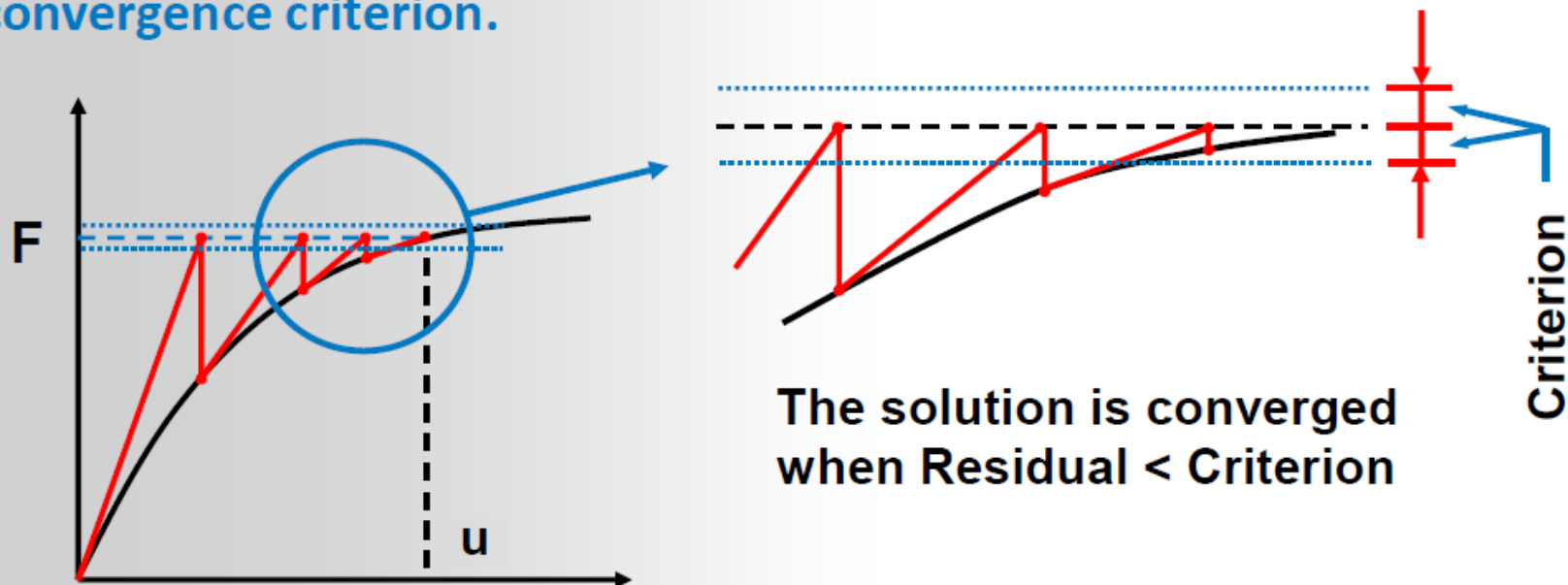
Details of "Analysis Settings"	
+ Step Controls	
+ Solver Controls	
+ Restart Controls	
- Nonlinear Controls	
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Off

... Obtaining a nonlinear solution

Recall that the Newton-Raphson method iterates to a converged solution using the equation $[K^T]\{\Delta u\} = \{F\} - \{F^{nr}\}$

The program solves this equation repeatedly until the residual (force imbalance), $\{F\} - \{F^{nr}\}$, becomes acceptably small.

The largest acceptable value for the residual is called the **force convergence criterion**.



... Obtaining a nonlinear solution

Expressed mathematically:

If $||\{R\}|| < (\epsilon_R R_{ref})$ Then the solution is converged.

Where:

- $\{R\} = \{F_a\} - \{F_{nr}\}$ is the residual vector
- $||\{R\}|| = (\sum R_i^2)^{1/2}$, is a vector **norm** of the residual (A norm is an operator that reduces a vector to a single scalar value)
 - $(\epsilon_R R_{ref})$ is the Force convergence criterion
 - ϵ_R is a tolerance factor and R_{ref} is a reference force value
 - R_{ref} is the norm of all applied forces and reactions, $||\{F\}||$ (automatically scaling the criterion to the magnitude of load)

Nonlinear Controls (Cont'd)

- In addition to force balance, a moment balance will also be included if rotational degrees of freedom (DOF) are present in the model (i.e. when beam and/or shell elements are present for example).

```

1      849  SHELL181      0.031  0.000037
2      849  SURF154      0.031  0.000037
*** LOAD STEP      1  SUBSTEP      1  COMPLETED.      CUM ITER =      11
*** TIME = 0.100000      TIME INC = 0.100000
*** AUTO STEP TIME:  NEXT TIME INC = 0.10000      UNCHANGED

FORCE CONVERGENCE VALUE = 37.86      CRITERION= 0.9880
MOMENT CONVERGENCE VALUE = 0.3632E-01 CRITERION= 5.960
EQUIL ITER      1  COMPLETED.  NEW TRIANG MATRIX.  MAX DOF INC= -1.426
FORCE CONVERGENCE VALUE = 5732.      CRITERION= 1.942
MOMENT CONVERGENCE VALUE = 461.1      CRITERION= 11.73

```

Static Structural (A5)

- Analysis Settings
- Fixed Support
- Pressure
- Solution (A6)
 - Solution Information
 - Directional Deformation
 - Contact Tool
 - Status
 - Pressure
 - Penetration
 - Gap

Details of "Analysis Settings"

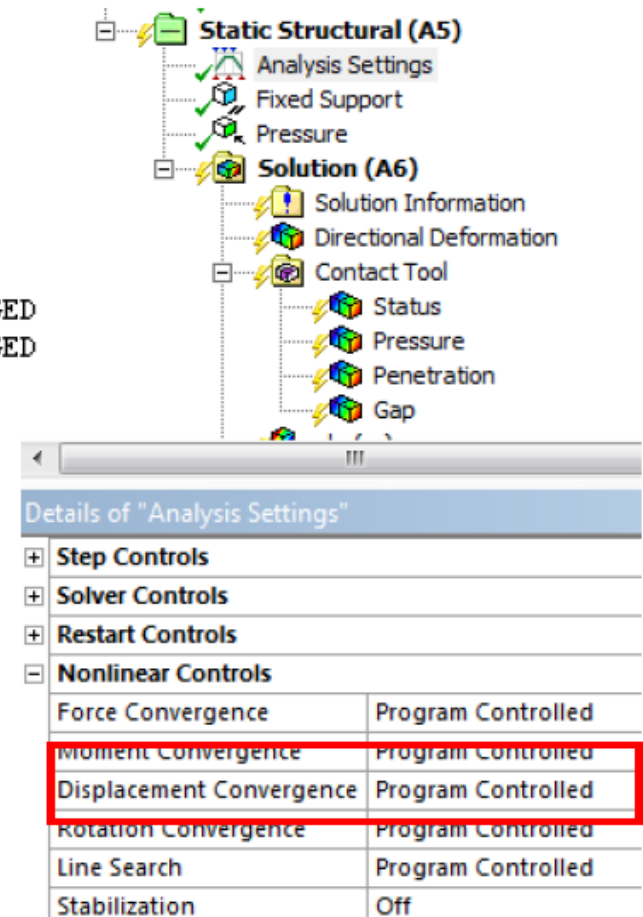
Step Controls	
Solver Controls	
Restart Controls	
Nonlinear Controls	
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Off

Nonlinear Controls (cont'd)

- Balance checks on displacement and/or rotational DOF values can also be added as a supplement to force/moment balances.

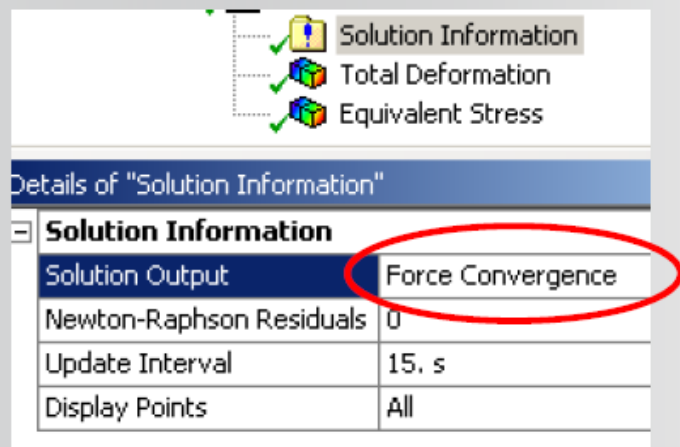
```
DISP CONVERGENCE VALUE    = 0.5294E-03  CRITERION= 0.1000E-01 <<< CONVERGED
ROT CONVERGENCE VALUE     = 0.3671E-05  CRITERION= 0.4957E-03 <<< CONVERGED
```

- When Joints are present in a model, these additional constraints will be added automatically.
- When nonlinear contact is present, these supplemental checks can sometimes be overly restrictive and can cause unnecessary divergence. User can remove as necessary.

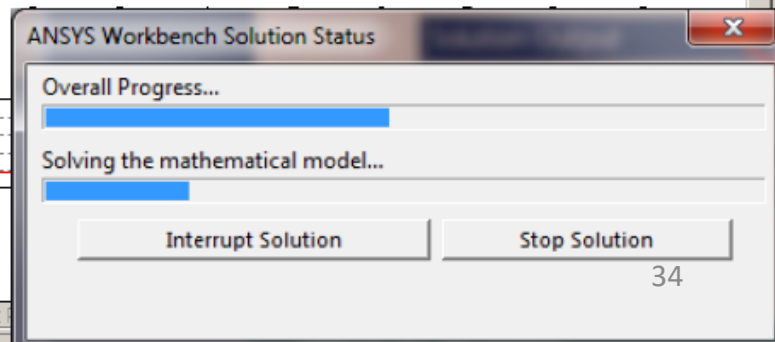
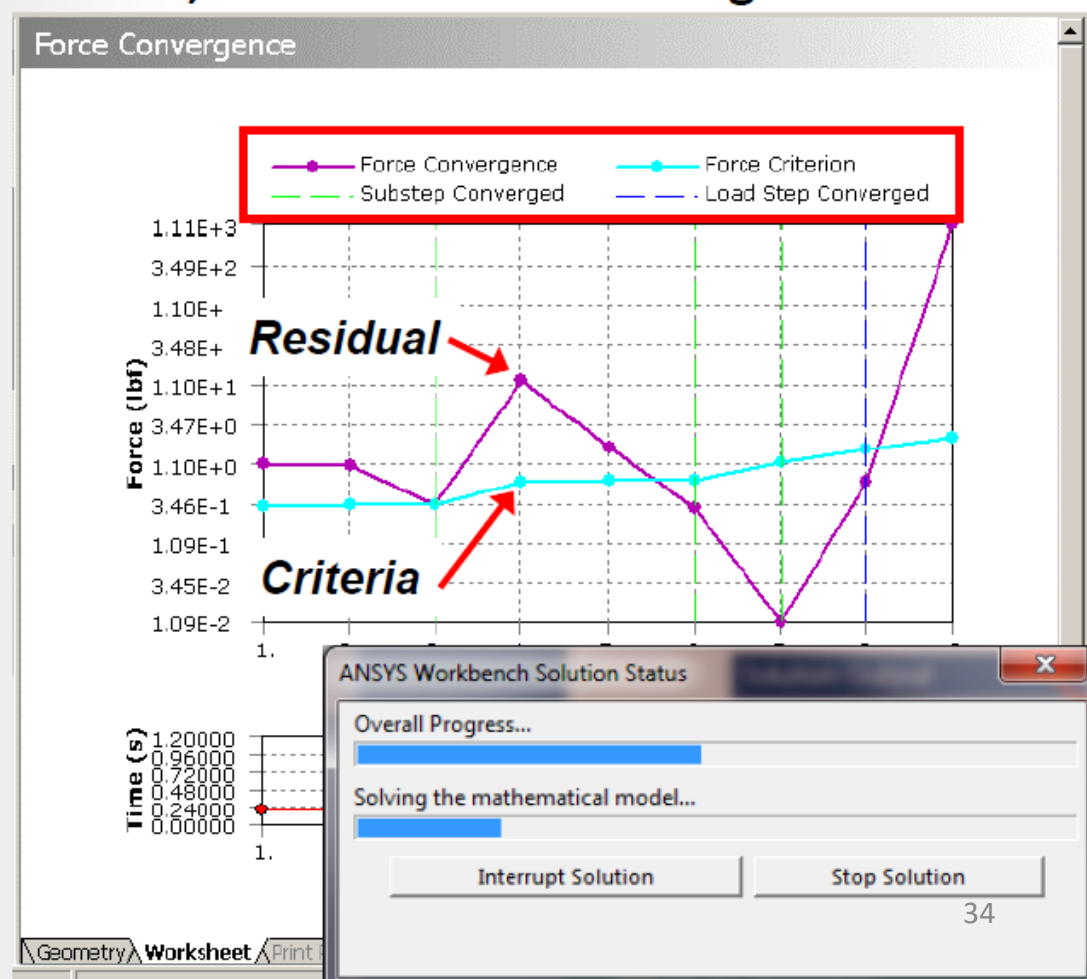


... Obtaining a nonlinear solution

- The Force Convergence graph displays a plot of the force criterion and residual forces ("force convergence") vs iteration.
- When the residual is less than the criterion, the solution is converged.

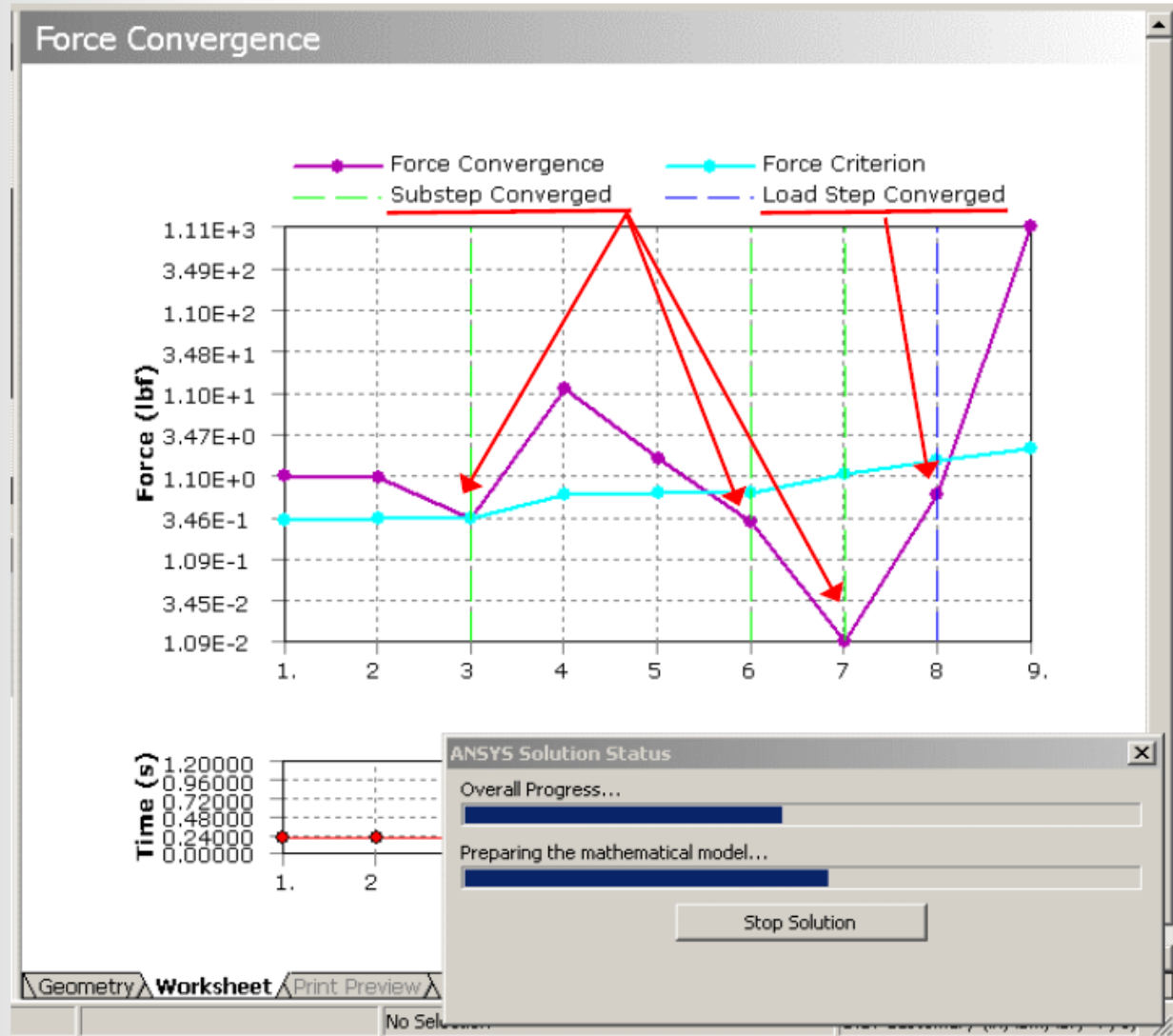


- Similar plots are available for moment convergence and for displacement and rotational DOF convergence when applicable.



... Obtaining a nonlinear solution

- Each converged substep is highlighted on this Force Convergence Graph with a vertical green dotted line.
- Each converged loadstep is highlighted with a blue dotted line.



... Obtaining a nonlinear solution

Project

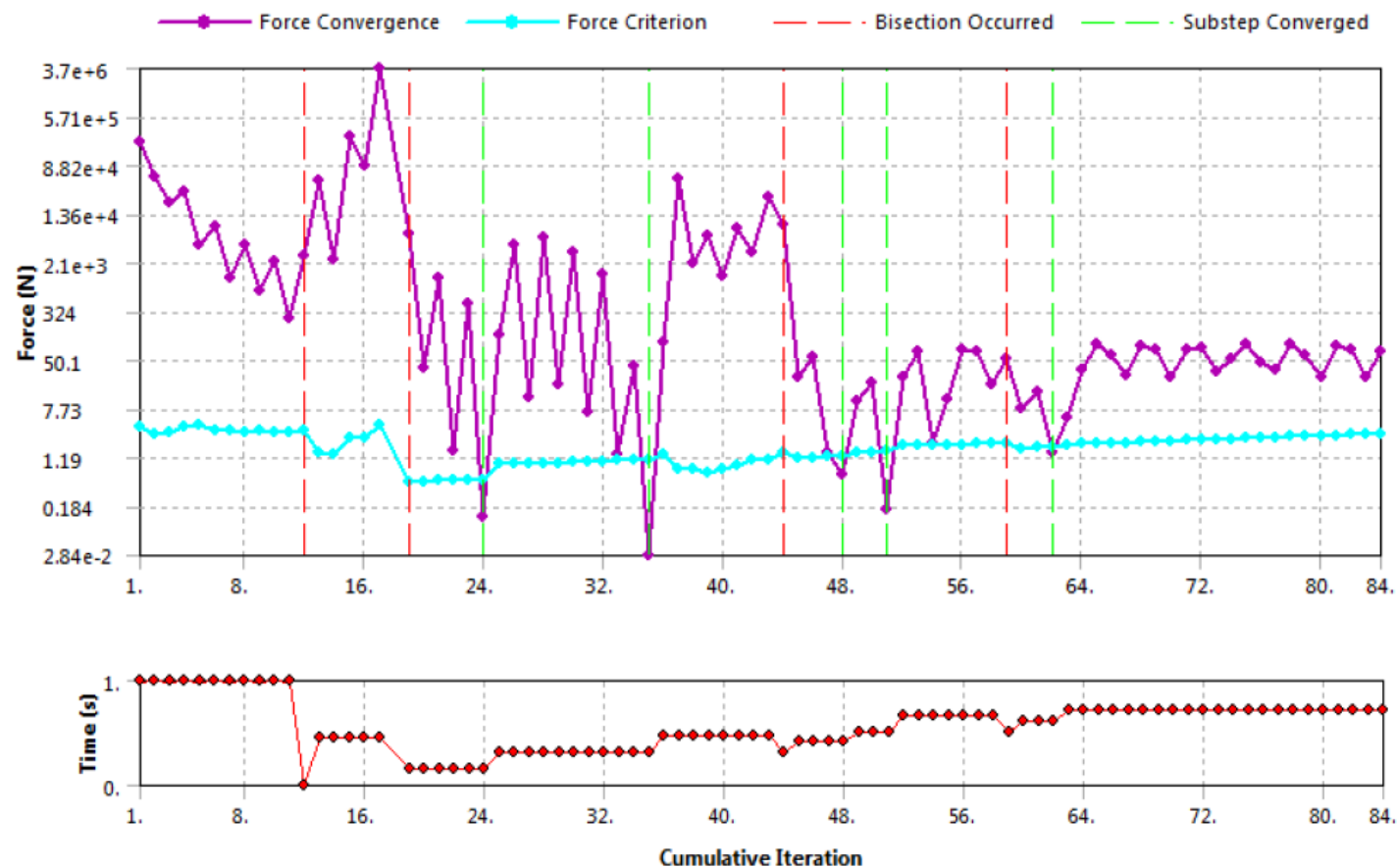
- Model (B4)
 - Geometry
 - Coordinate Systems
 - Mesh
 - Large Deflection-Linear Mat'l (B5)
 - Analysis Settings
 - Fixed Support
 - Force
 - Solution (B6)
 - Solution Information
 - Total Deformation
 - Equivalent Stress
 - Force Reaction

Details of "Solution Information"

Solution Information	
Solution Output	Force Convergence...
Newton-Raphson Residuals	0
Update Interval	2.5 s
Display Points	All

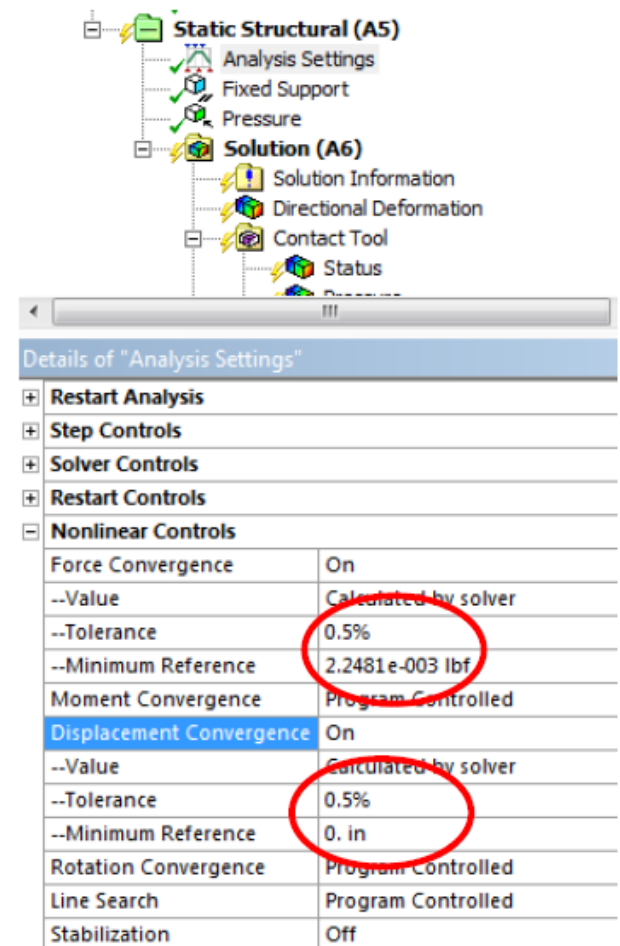
Worksheet

Force Convergence



Nonlinear Controls (cont'd)

- If you add any convergence criteria, the program deletes *all* the default criteria!
- For example, if you override program control by adding a displacement convergence check, the force convergence check will be deleted.
- *Make sure you reestablish the force convergence check.*
- After redefining convergence criteria, you should always confirm the specifications reported in the Solution Information branch to ensure intended balance checks are active.



CONVERGENCE ON F BASED ON THE NORM OF THE N-R LOAD
 WITH A TOLERANCE OF 0.5000E-02 AND A MINIMUM REFERENCE VALUE OF 0.2248E-02
 USING THE L2 NORM (CHECK THE SRSS VALUE)

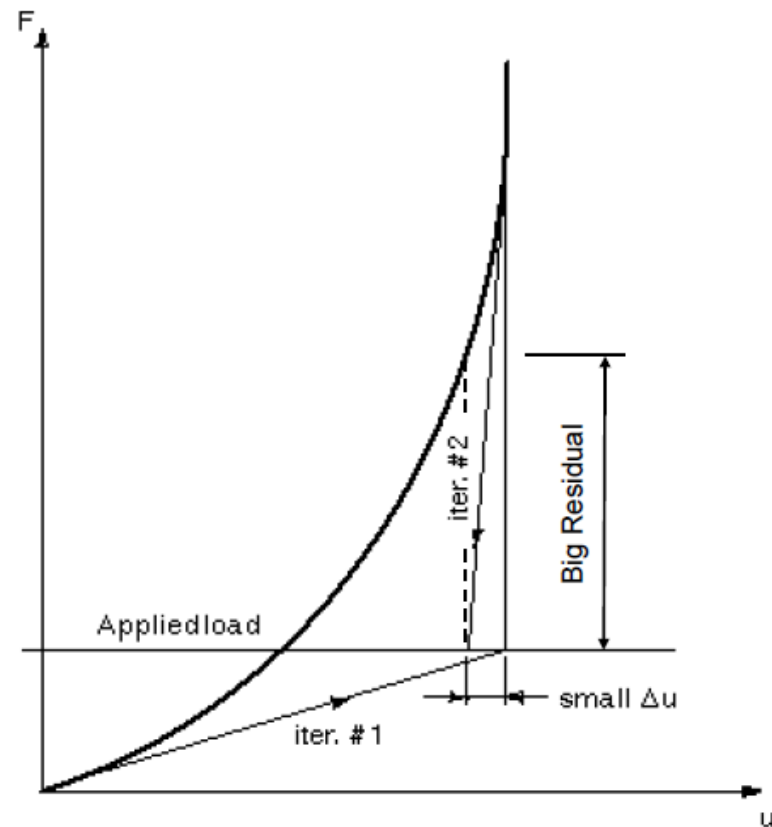
CONVERGENCE ON U BASED ON THE NORM OF THE TOTAL
 WITH A TOLERANCE OF 0.5000E-02 AND A MINIMUM REFERENCE VALUE OF 0.000
 USING THE L2 NORM (CHECK THE SRSS VALUE)

... Obtaining a nonlinear solution

Nonlinear Controls (cont'd)

Why must you re-establish a force convergence criterion?

- Because displacement-based checking is a *relative* measure of convergence, it should only be used as a supplement to force-based convergence.
- Force-based convergence provides an *absolute* measure of convergence, as it is a measure of equilibrium between the internal and external forces.



Relying on displacement convergence alone can in some cases lead to erroneous results.

... Obtaining a nonlinear solution

Nonlinear Controls (cont'd)

- The Minimum reference value (MINREF) is a safety feature that prevents your solution from trying to converge to a zero tolerance.
 - If free-body (unconstrained) systems or mechanisms have no external forces, the criterion ($e_R * ||\{F\}||_2$) will be zero. If the criterion is zero, the solution will *never* converge!
 - In such cases, the program redefines the criterion to be ($e_R * \text{MINREF}$). Where e_R is the convergence tolerance value.
 - The default value that WB-Mechanical uses for MINREF depends on the physics of the problem.

☐ Nonlinear Controls	
Force Convergence	On
--Value	Calculated by solver
--Tolerance	0.5%
--Minimum Reference	2.2481e-003 lbf

... Obtaining a nonlinear solution

Convergence criteria guidelines:

- Default convergence criteria work well most of the time.
 - You should rarely need to change the criteria.
- To tighten or loosen a criterion, don't change the default reference value, but instead change the tolerance factor by one or two orders of magnitude.
- Do not use a “loose” criterion to eliminate convergence difficulties.
 - This simply allows the solution to “converge” to an incorrect result!
- Tightening the criterion requires more equilibrium iterations.
- Review any MINREF warning messages during solution. Make sure the minimum reference value used makes sense for the problem being solved.

... Obtaining a nonlinear solution

Nonlinear Controls (cont'd)

- Stabilization is a nonlinear control intended to deal with structural instability (buckling and/or localized yielding).
 - Analogous to adding artificial dampers or dashpot elements at strategic locations.

Refer to Chapter 5 for detailed discussion.

Details of "Analysis Settings"	
+ Restart Analysis	
+ Step Controls	
+ Solver Controls	
+ Restart Controls	
- Nonlinear Controls	
Force Convergence	On
--Value	Calculated by solver
--Tolerance	0.5%
--Minimum Reference	2.2481e-003 lbf
Moment Convergence	Program Controlled
Displacement Convergence	On
--Value	Calculated by solver
--Tolerance	0.5%
--Minimum Reference	0. in
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Off
+ Output Controls	Off
+ Analysis Data Management	Constant Reduce
+ Visibility	

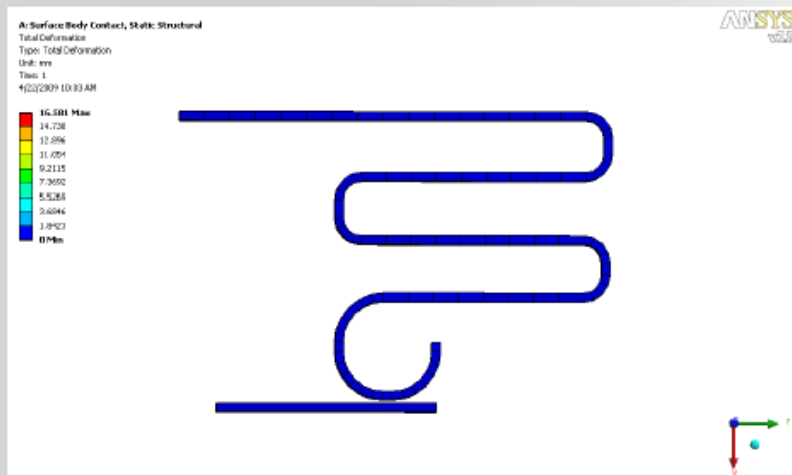
C. Reviewing nonlinear results

What is different about reviewing nonlinear results?

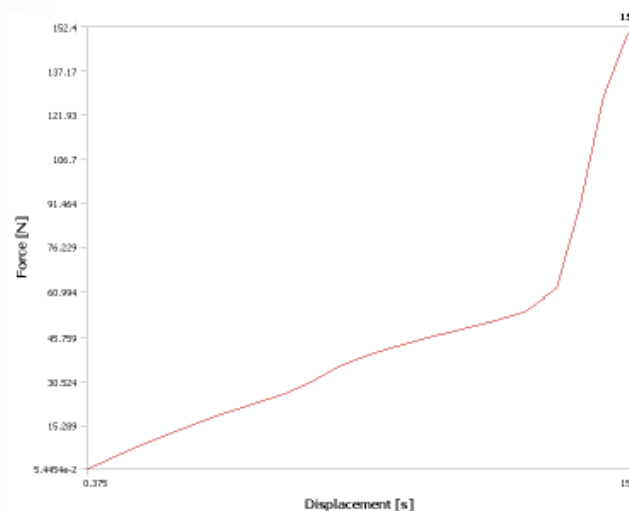
The procedure for reviewing nonlinear results is similar to that of a linear problem. The difference is that there is usually more information to process

- multiple results sets
- more information per result set (i.e. contact status, pressure, penetration, inelastic strains due to plasticity and or creep,...etc).

A nonlinear analyses produces a *response history*



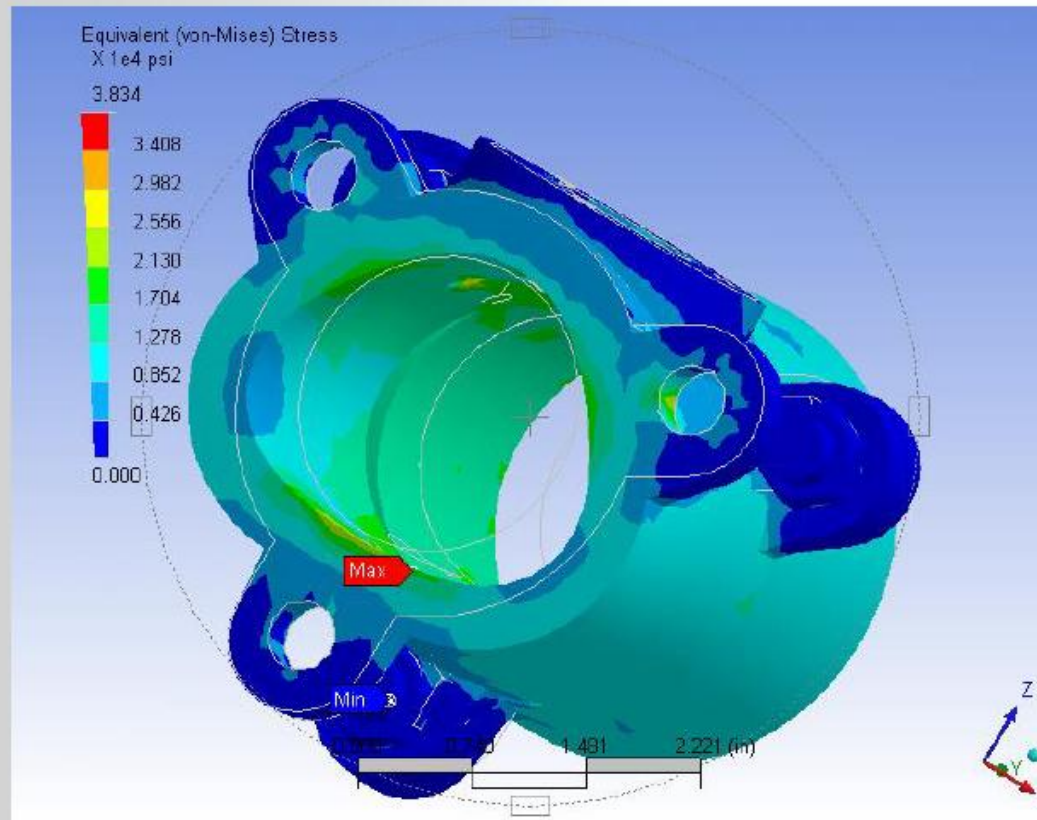
Animated response history



Response history graph

... Reviewing nonlinear results

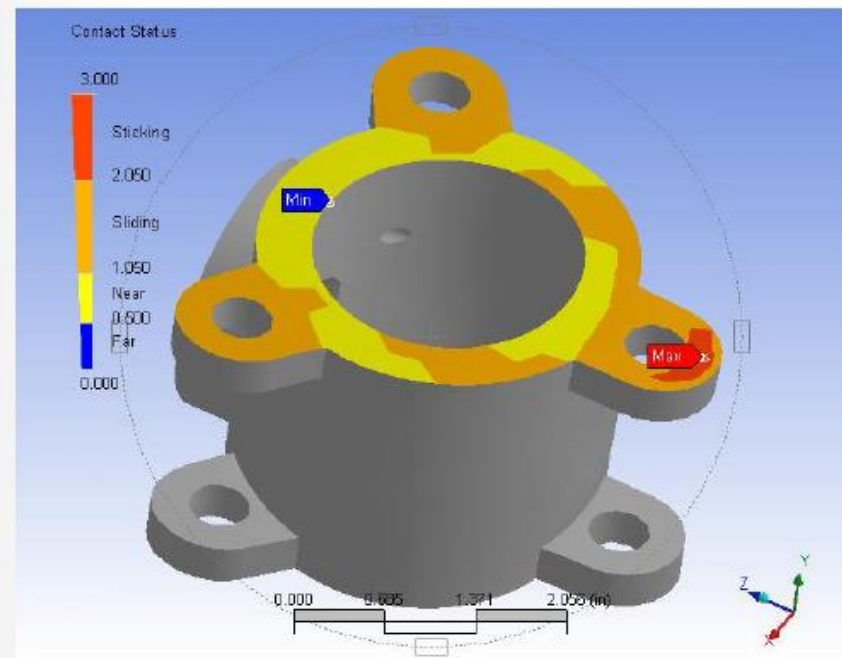
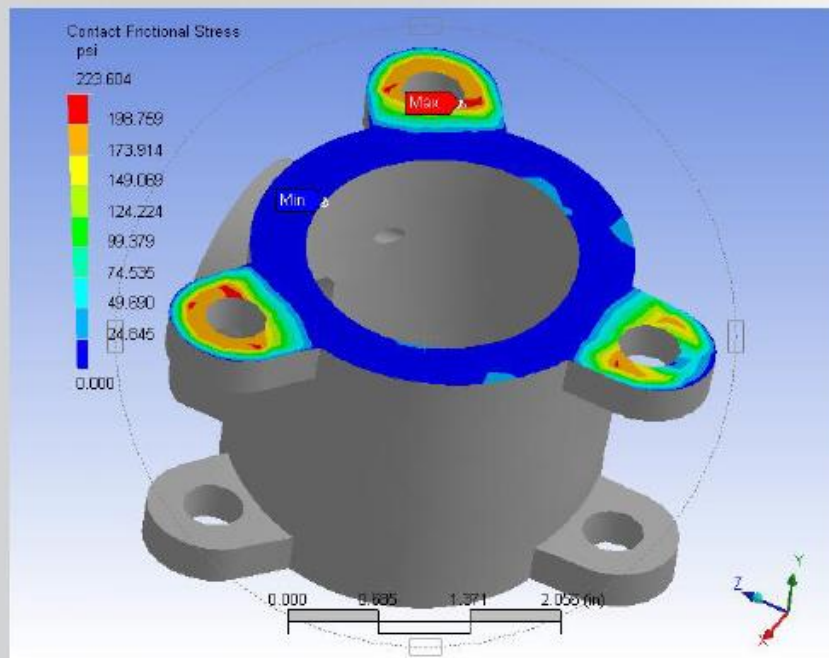
- In large deformation problems, one usually should view the deformation with “Actual” scaling from the Result toolbar
- Any of the structural results may be requested, such as Equivalent Stress, shown below



Model shown is from a sample Unigraphics assembly.

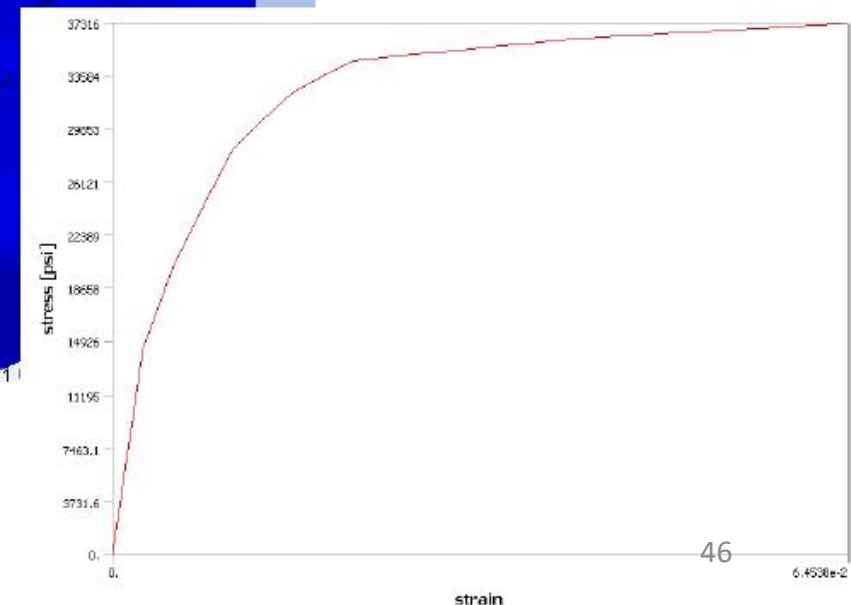
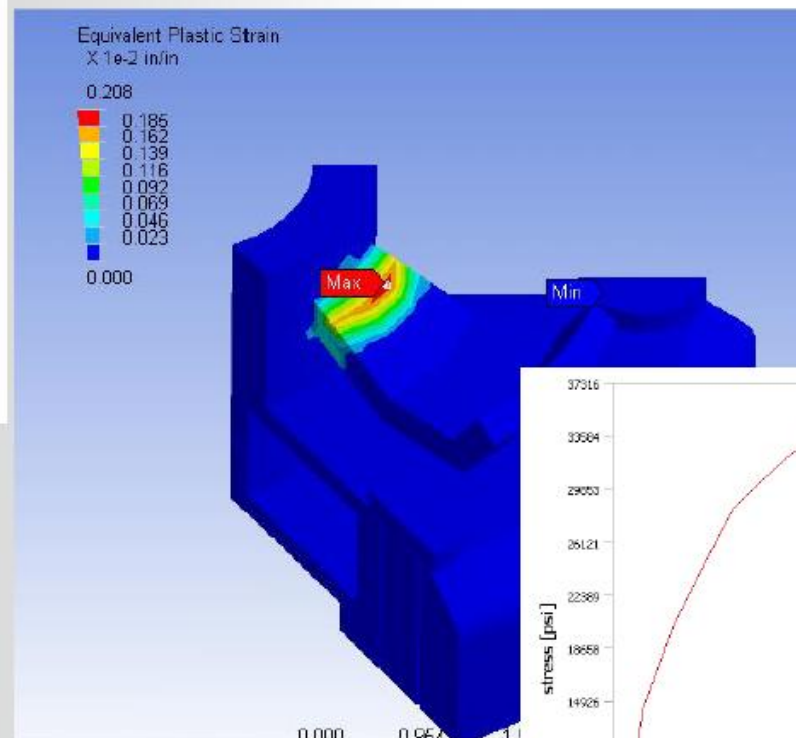
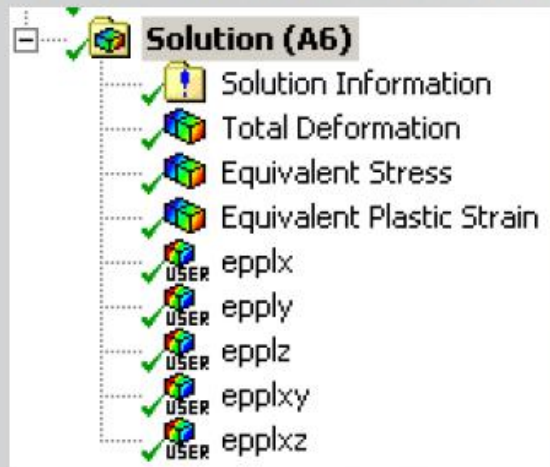
... Reviewing nonlinear results

- If contact is defined, a contact tool can be used to postprocess contact related results (pressure, penetration, frictional stress, status,..etc)
- We can explore this tool in greater detail in Chapter 3



... Reviewing nonlinear results

- If nonlinear material is defined, various stress and strain components can be requested.
- We will explore this in greater detail in Chapter 4.



Workshop 2B – Restart Control

Please refer to your *Workshop Supplement* for instructions on:

Workshop 2B- Restart Control

Workshop 2B: Performing a Restart

Goal

Use Restart Control technology to remedy a nonconverged solution and add a new load step

Model Description

- Same as Workshop 2A

